Exploratory Analysis on Crime Records in Los Angeles

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Repository Link

Section 1, DATASCI 200
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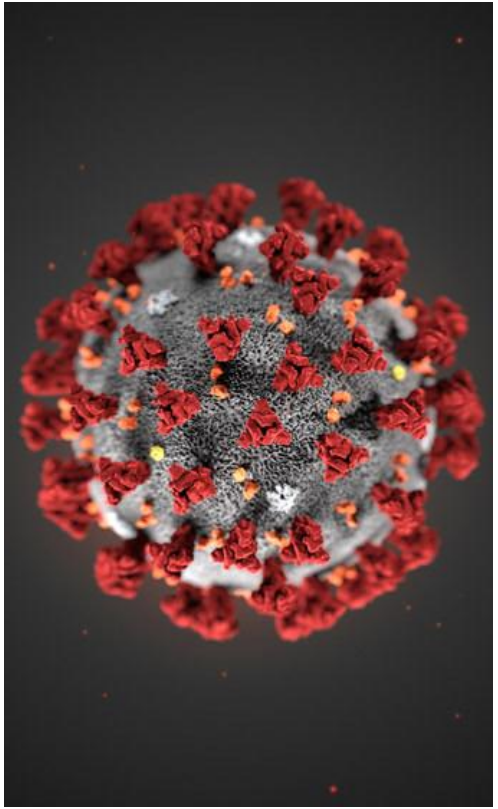
Introduction

COVID was a global time of change. Five years later, peculiarities from our shared experiences of the pandemic seem surreal.

With so many aspects of our lives changed, our team set out to investigate how crime also changed during this period (March 2020 - March 2024).

We decided to focus our analysis on three Hypothesis:

1. Crime changed from COVID to Post COVID and also crime amount increased by area.
2. Crime changed in frequency, type, and time between the two periods.
3. Crime against certain victim demographics changed during these periods.



Exploratory Data Analysis

1. We made sure the dataframe matched our expectations
   a. 28 Columns and over 1M rows
   b. Majority of essential Data was filled out (fig 1)
   c. NaN values made sense based on the column type (fig 2)
2. Check Variables
   a. Victim Age
      i. About 134 entries less than 0
      ii. We found 260k entries with age equal 0
   b. Victim Sex
      i. Approximately 115 victims with undocumented label
      ii. Approximately 97k listed as “unknown”
   c. Duplicates
      i. About 3k
      ii. Dropped
3. Dataset Limitations
   a. Only 3 months of pre-COVID data
   b. 1 Year of Post-COVID data
      i. Data Normalization used for Comparisons (Mean or Percent)
   c. The Dataset changed reporting standards after Mar 2024
      i. Most Crime types began to be underreported after that period
      ii. We decided to exclude post Mar 2024 from our analysis.
4. During COVID: March 1, 2020 (US lockdown time) - May 31, 2023
   Post-COVID: June 1, 2023 (US unlock date) - March 31, 2024

Fig 1

Date Rptd	1005149
DATE OCC	1005149
TIME OCC	1005149
AREA	1005149
AREA NAME	1005149
Rpt Dist No	1005149
Part 1-2	1005149
Crm Cd	1005149
Crm Cd Desc	1005149
Mocodes	853408
Vict Age	1005149
Vict Sex	860384
Vict Descent	860372
Premis Cd	1005133
Premis Desc	1004561
Weapon Used Cd	327264
Weapon Desc	327264
Status	1005148
Status Desc	1005149
Crm Cd 1	1005138
Crm Cd 2	69153
Crm Cd 3	2314
Crm Cd 4	64
LOCATION	1005149
Cross Street	154240
LAT	1005149
LON	1005149
dtype: int64	

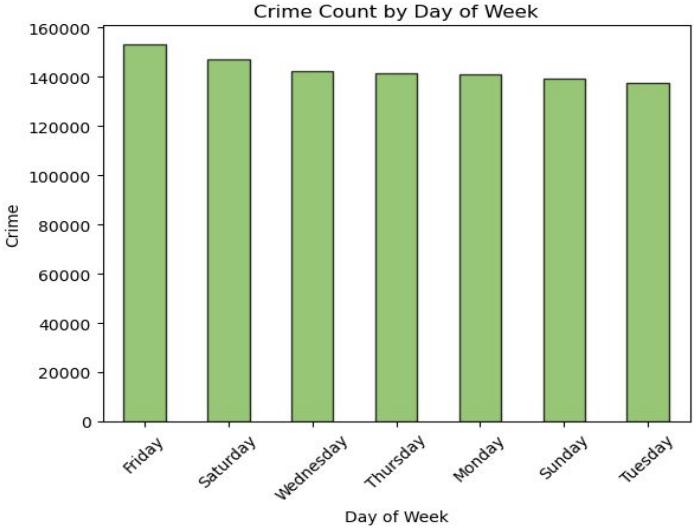
Fig 2

Date Rptd	0
DATE OCC	0
TIME OCC	0
AREA	0
AREA NAME	0
Rpt Dist No	0
Part 1-2	0
Crm Cd	0
Crm Cd Desc	0
Mocodes	151741
Vict Age	0
Vict Sex	144765
Vict Descent	144777
Premis Cd	16
Premis Desc	588
Weapon Used Cd	677885
Weapon Desc	677885
Status	1
Status Desc	0
Crm Cd 1	11
Crm Cd 2	935996
Crm Cd 3	1002835
Crm Cd 4	1005085
LOCATION	0
Cross Street	850909
LAT	0
LON	0
dtype: int64	

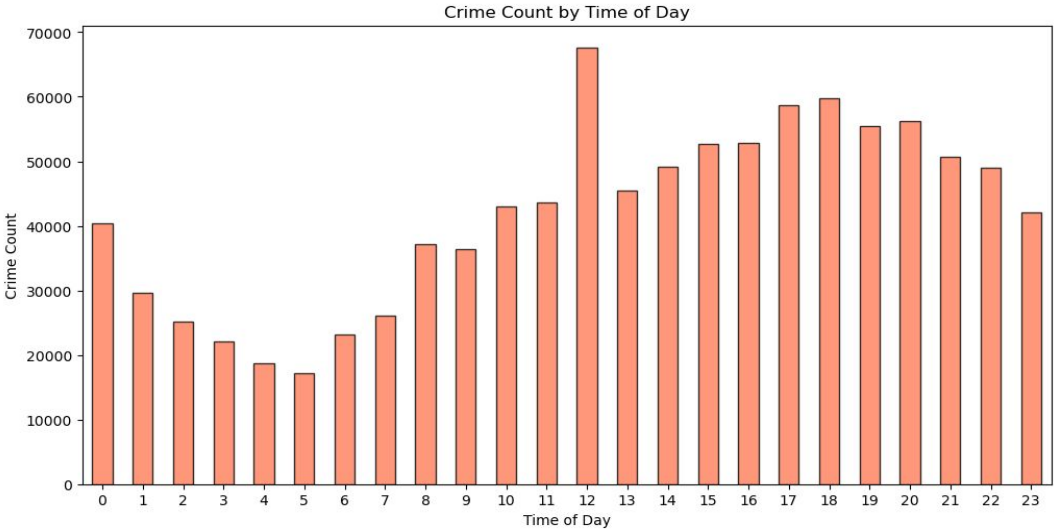
Exploratory Data Analysis Continued

Initial Plots: Crime as a function of Time

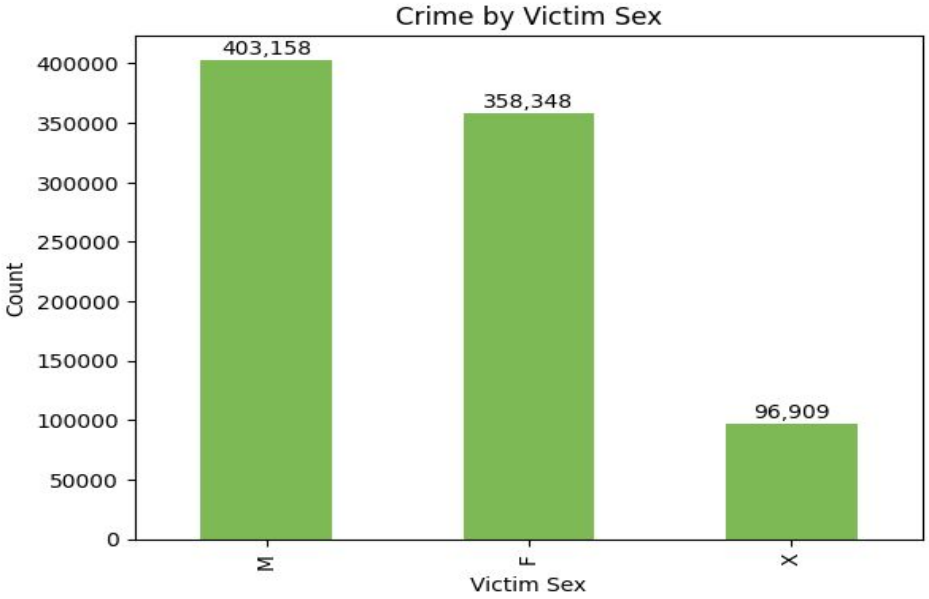
Day of Week doesn't seem to play a large role in when crime is conducted



Time of Day plays a much more important role in the conduct of time

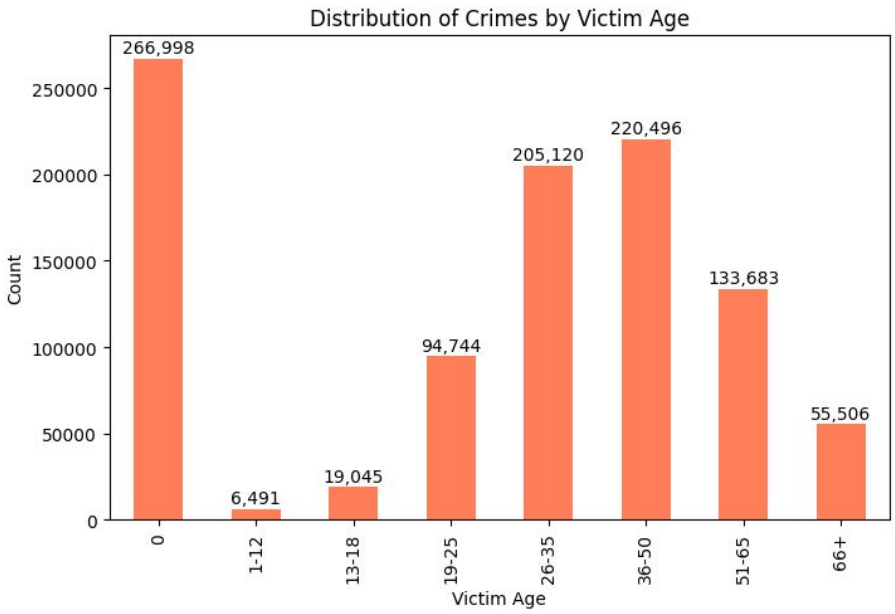


Exploratory Data Analysis Continued



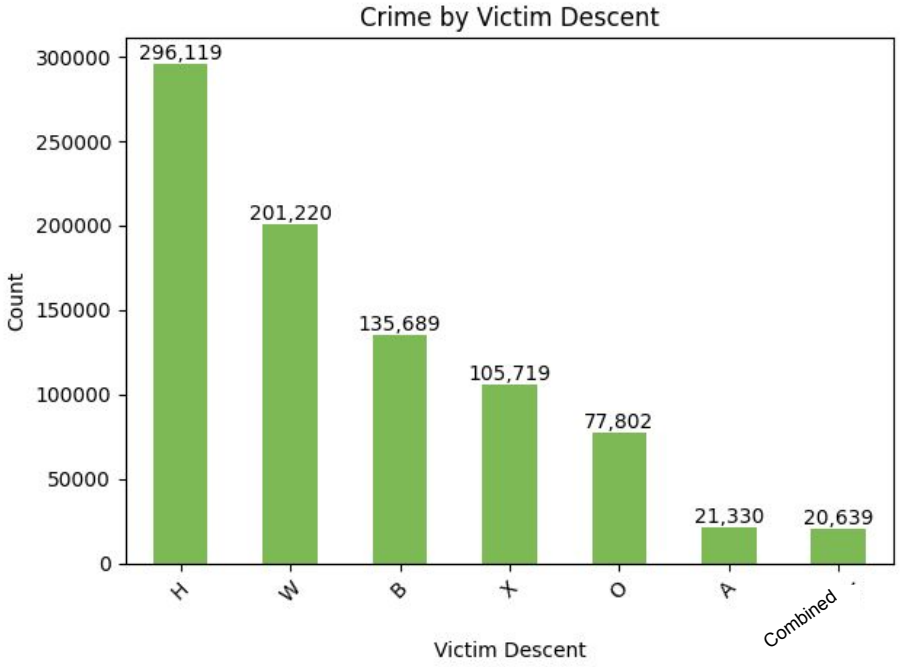
F - Female M - Male X - Unknown

Exploratory Data Analysis Continued




```
crm_cd_desc
VEHICLE - STOLEN                                                    113972
THEFT FROM MOTOR VEHICLE - PETTY ($950 & UNDER)                      24963
SHOPLIFTING - PETTY THEFT ($950 & UNDER)                             21410
BURGLARY                                                             18017
VANDALISM - FELONY ($400 & OVER, ALL CHURCH VANDALISMS)              14688
TRESPASSING                                                           9406
THEFT-GRAND ($950.01 & OVER)EXCPT,GUNS,FOWL,LIVESTK,PROD              7424
THEFT PLAIN - PETTY ($950 & UNDER)                                    6764
ROBBERY                                                               6495
VANDALISM - MISDEAMEANOR ($399 OR UNDER)                              5291
SHOPLIFTING-GRAND THEFT ($950.01 & OVER)                              3946
EMBEZZLEMENT, GRAND THEFT ($950.01 & OVER)                            3525
OTHER MISCELLANEOUS CRIME                                             2559
DISCHARGE FIREARMS/SHOTS FIRED                                        2135
VEHICLE, STOLEN - OTHER (MOTORIZED SCOOTERS, BIKES, ETC)              2100
BURGLARY FROM VEHICLE                                                 1956
ASSAULT WITH DEADLY WEAPON, AGGRAVATED ASSAULT                        1932
THEFT FROM MOTOR VEHICLE - GRAND ($950.01 AND OVER)                   1788
BATTERY POLICE (SIMPLE)                                               1688
FAILURE TO YIELD                                                      1270
THEFT OF IDENTITY                                                     1174
BURGLARY, ATTEMPTED                                                   1154
ARSON                                                                 1090
DOCUMENT FORGERY / STOLEN FELONY                                       941
...
RAPE, ATTEMPTED                                                          1
TELEPHONE PROPERTY - DAMAGE                                              1
CONTRIBUTING                                                             1
PURSE SNATCHING - ATTEMPT                                                1
Name: count, dtype: int64
```

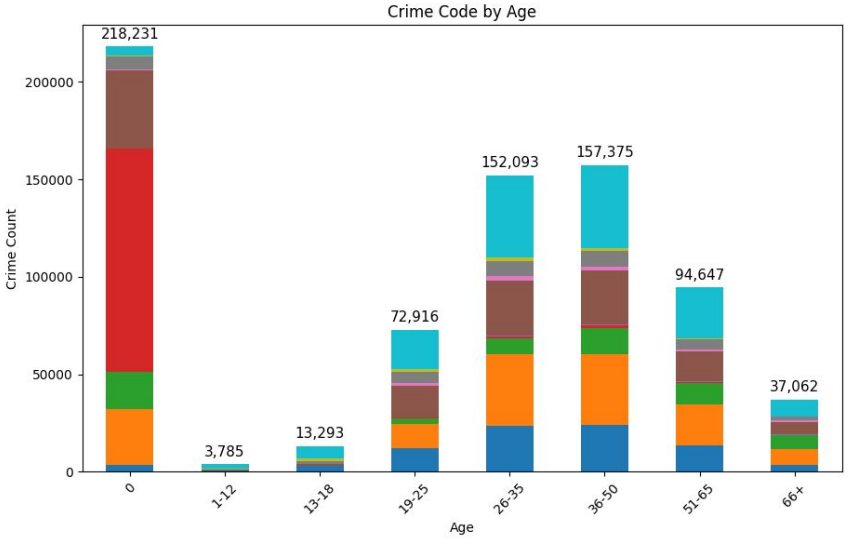
Exploratory Data Analysis Continued



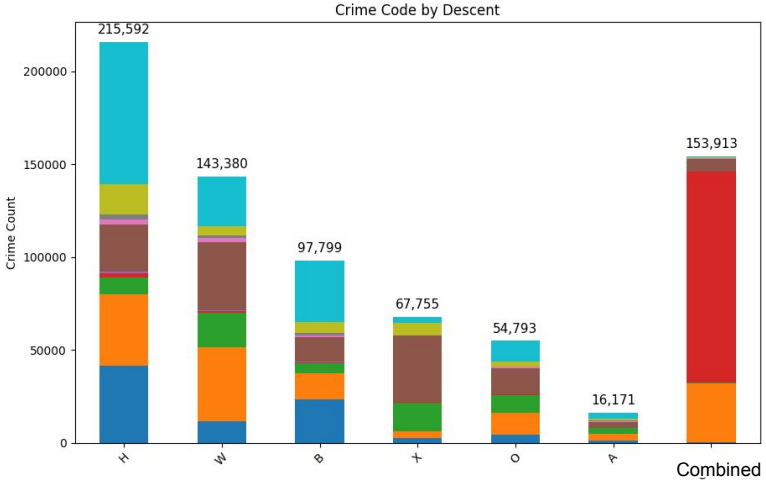
- H - Hispanic/Latin/Mexican
- W - White
- B - Black
- X - Unknown
- A - Other Asian
- O-Other
- Combined (combines the rest)

<u>Combined Groups:</u> C - Chinese D - Cambodian F - Filipino G - Guamanian I - American Indian/Alaskan Native J - Japanese K - Korean L - Laotian P - Pacific Islander S - Samoan U - Hawaiian V - Vietnamese Z - Asian Indian

Demographics By Crime Code



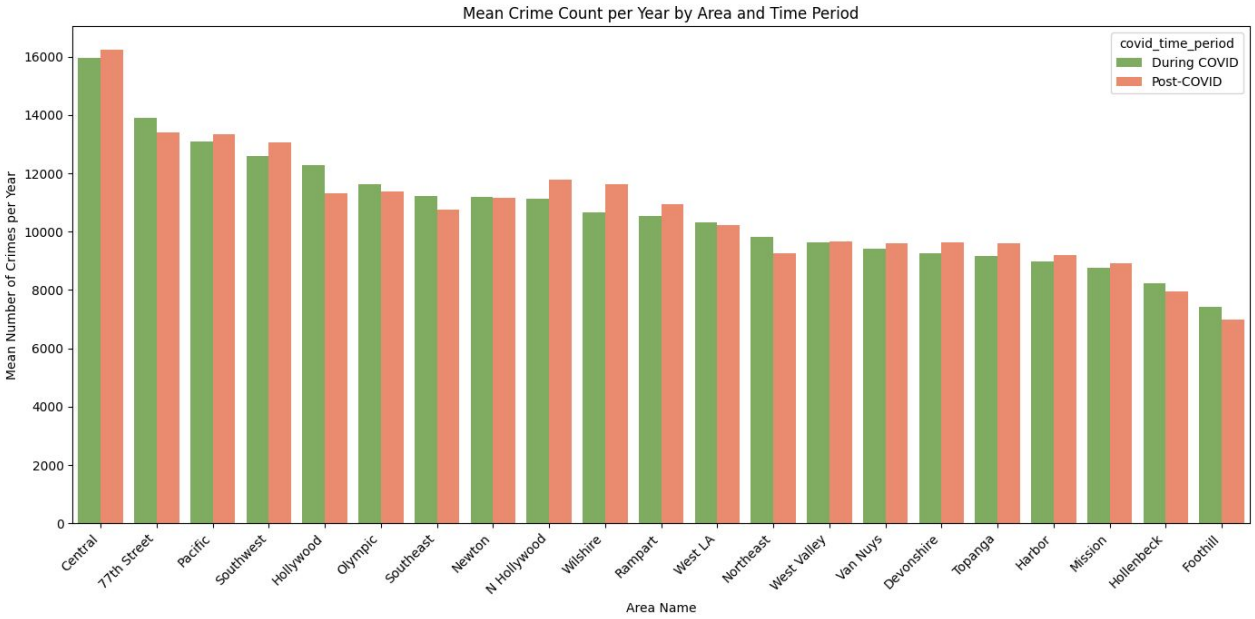
Top Crimes: simple assault, burglary theft from vehicles, and other theft



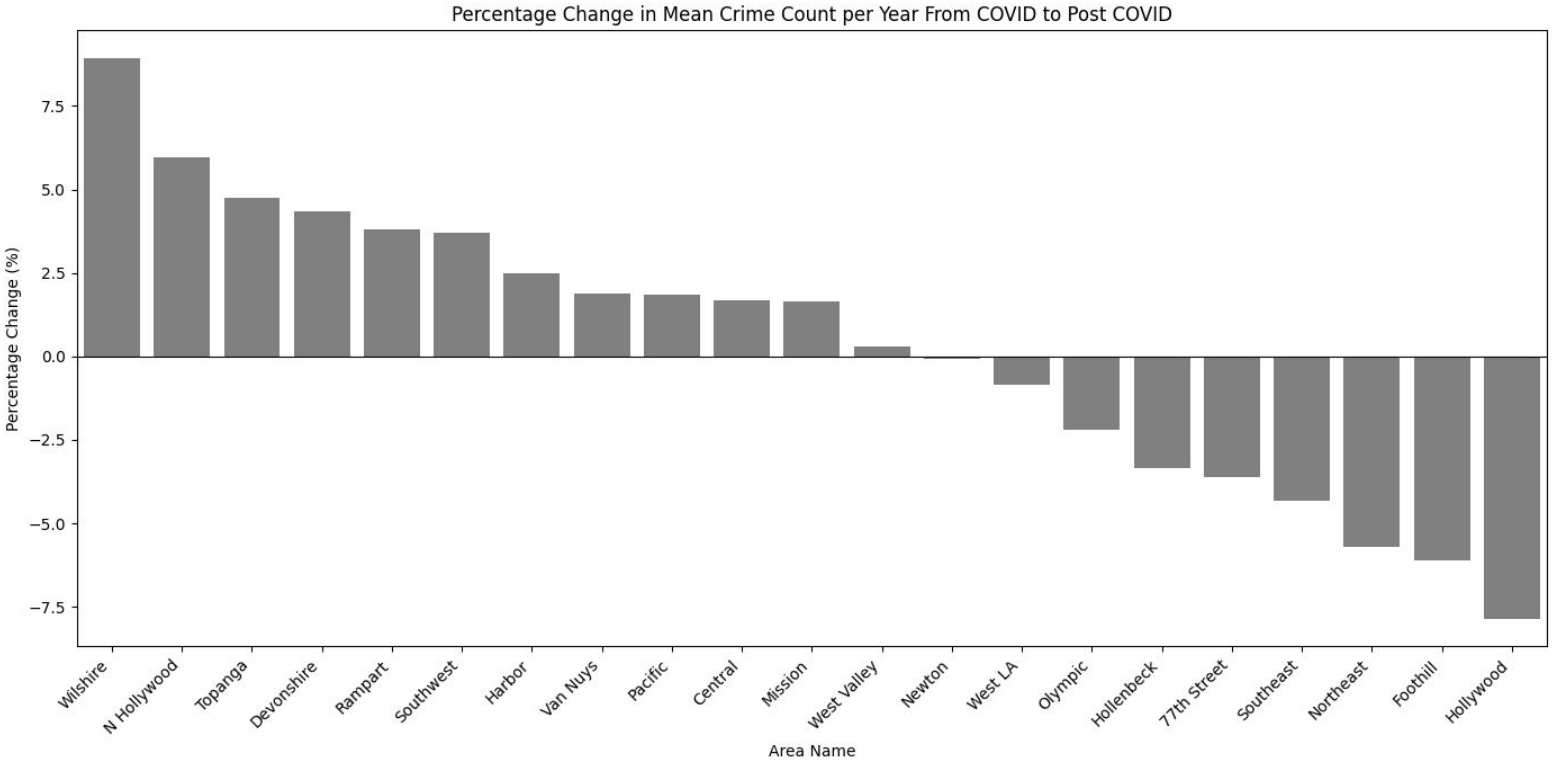
Hypothesis 1:
We assume that crime increase from COVID to Post COVID and also crime amount changed by area

Primary Questions

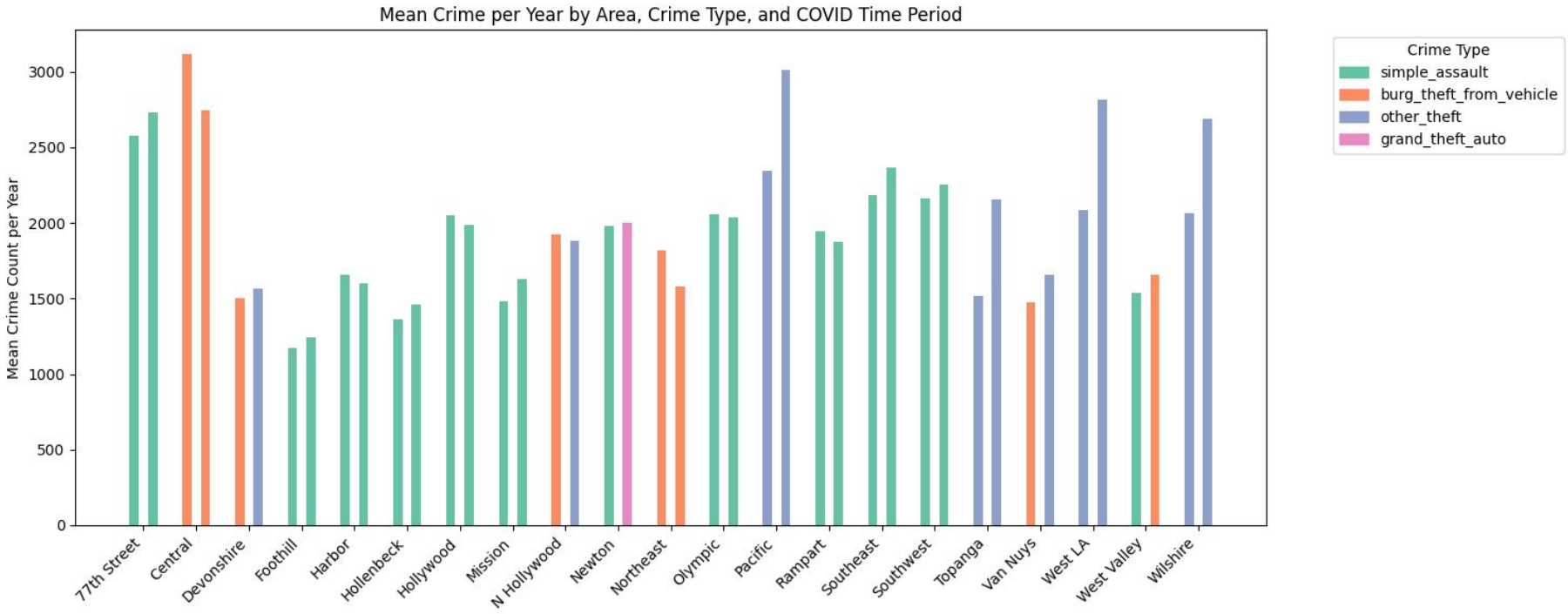
Yes, Crime amount changed by area during the COVID and Post-COVID periods



No, not all the areas have the crime percentage increased, only half of the area



The Dominant Crime Types Shifts in 5 areas, not all crime amount changed by area



Question 2: How did crime change between the two periods?

Did crime change in frequency, type, or time?

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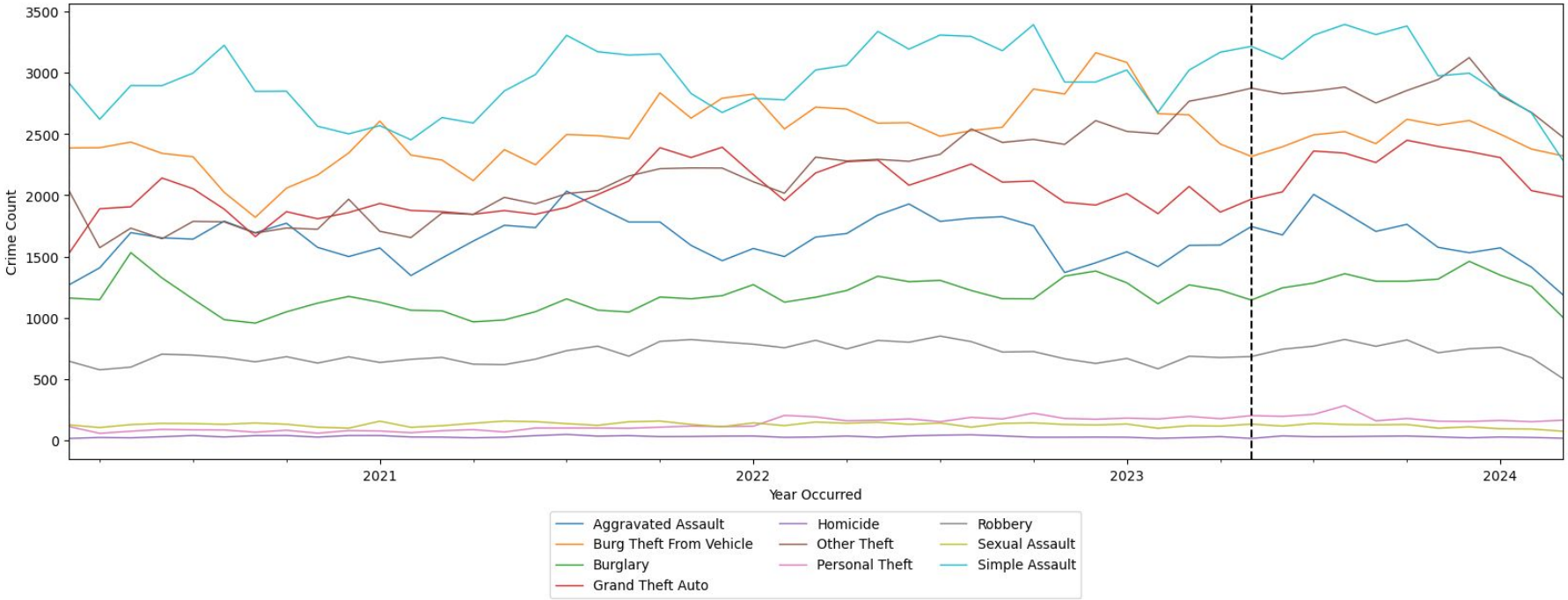
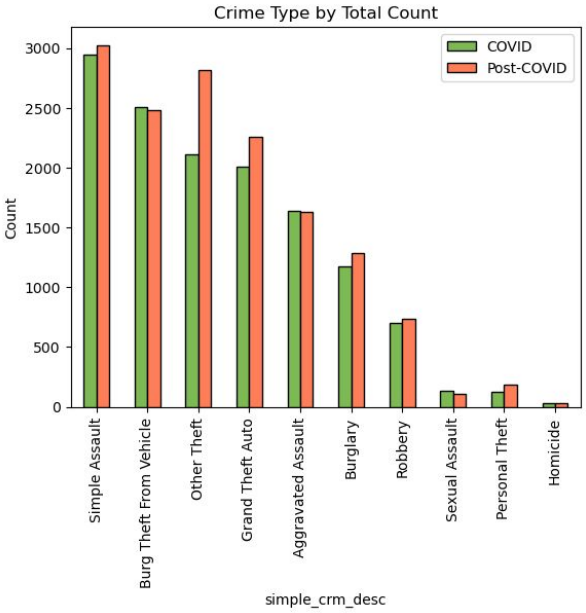
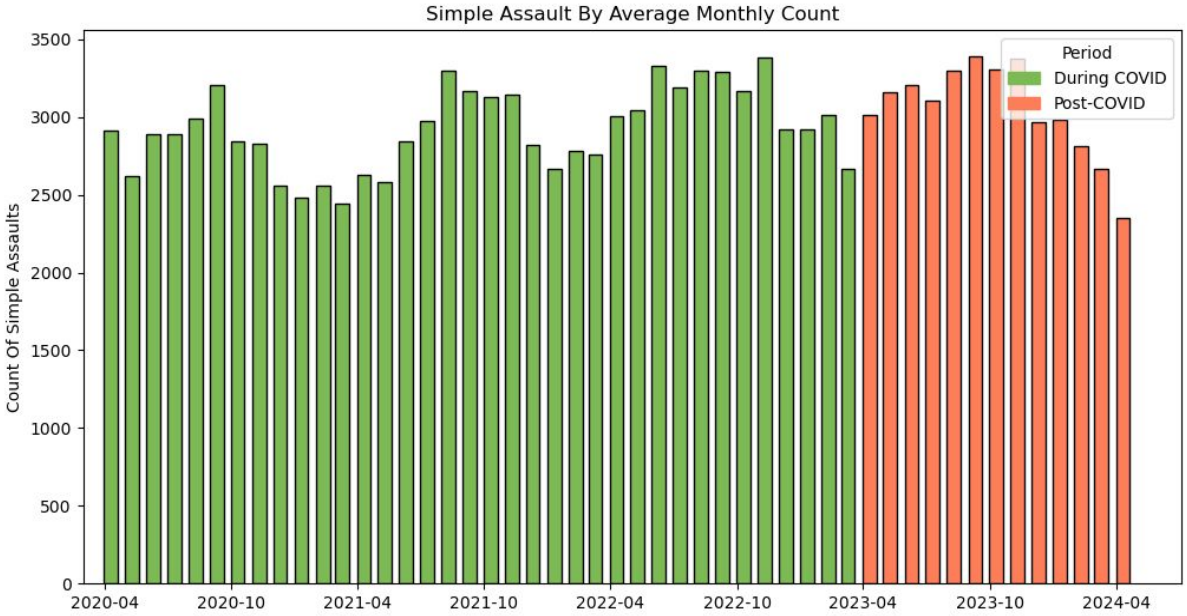
A quick snapshot of Major Crime Types over the Period



Question 2: How did crime change between the two periods? Did crime change in frequency, type, or time?

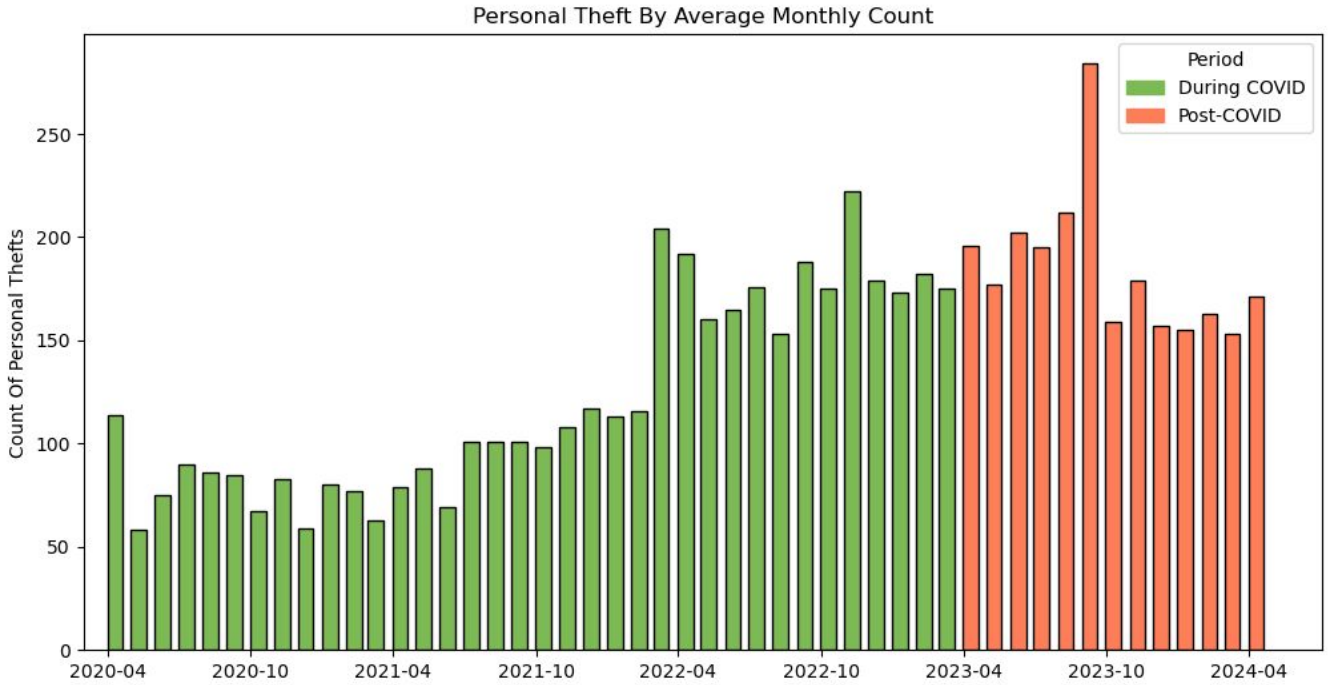
Many types of crime were similar through both periods.





Question 2: How did crime change between the two periods? Did crime change in frequency, type, or time?

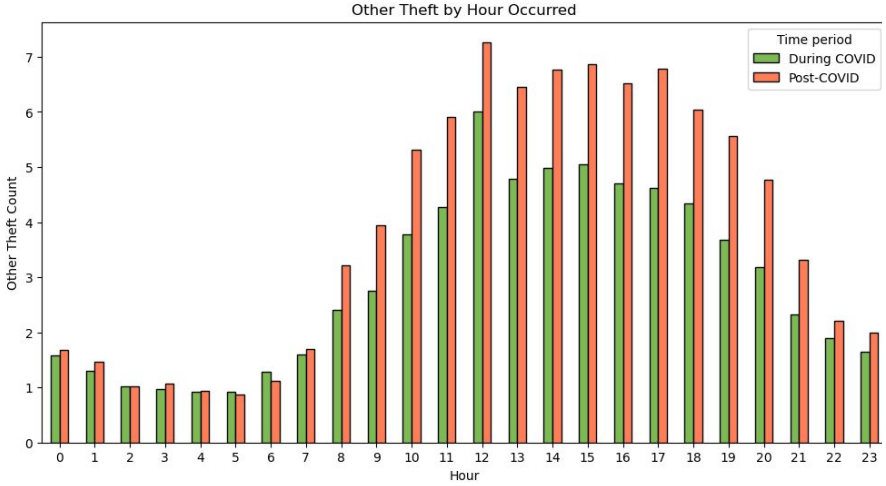
Other Types of Crime appear to have been heavily repressed by COVID, except for Personal Theft



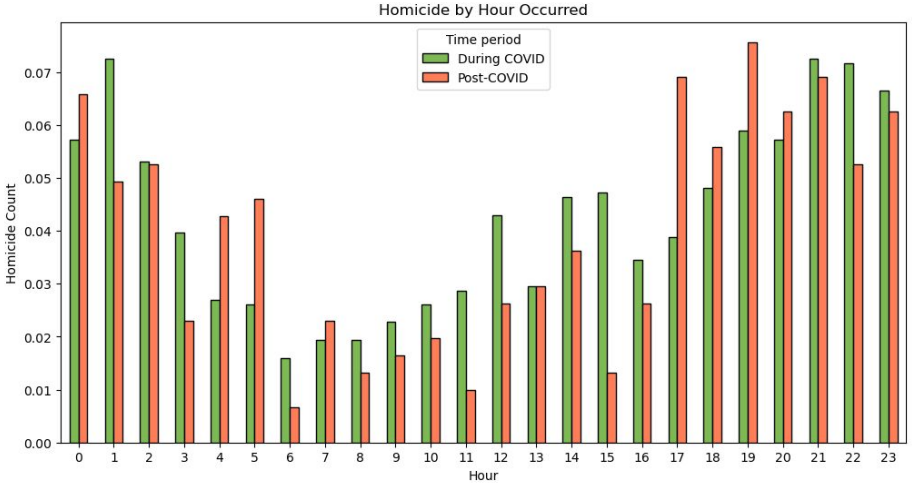
Question 2: How did crime change between the two periods? Did crime change in frequency, type, or time?

Time also played an important role.

"Other Theft" merely increased in frequency during normal hours.

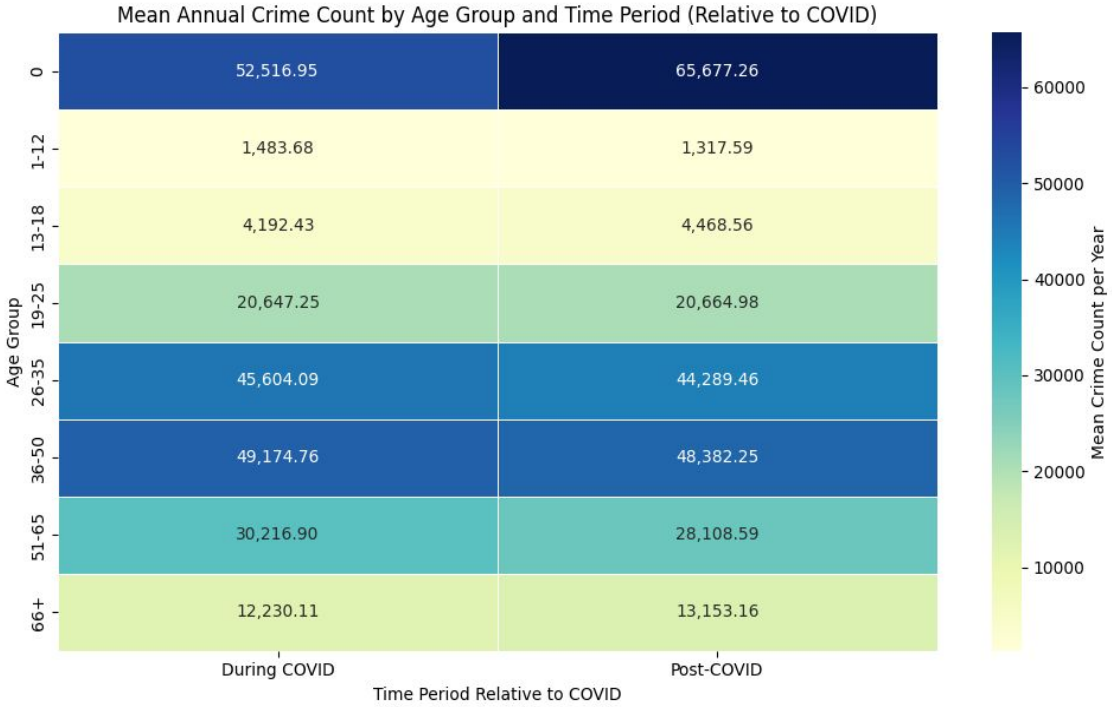


While Homicide changed in time of occurrence.



Hypothesis 3:
How did crime against certain victim demographics change during COVID versus post- COVID?

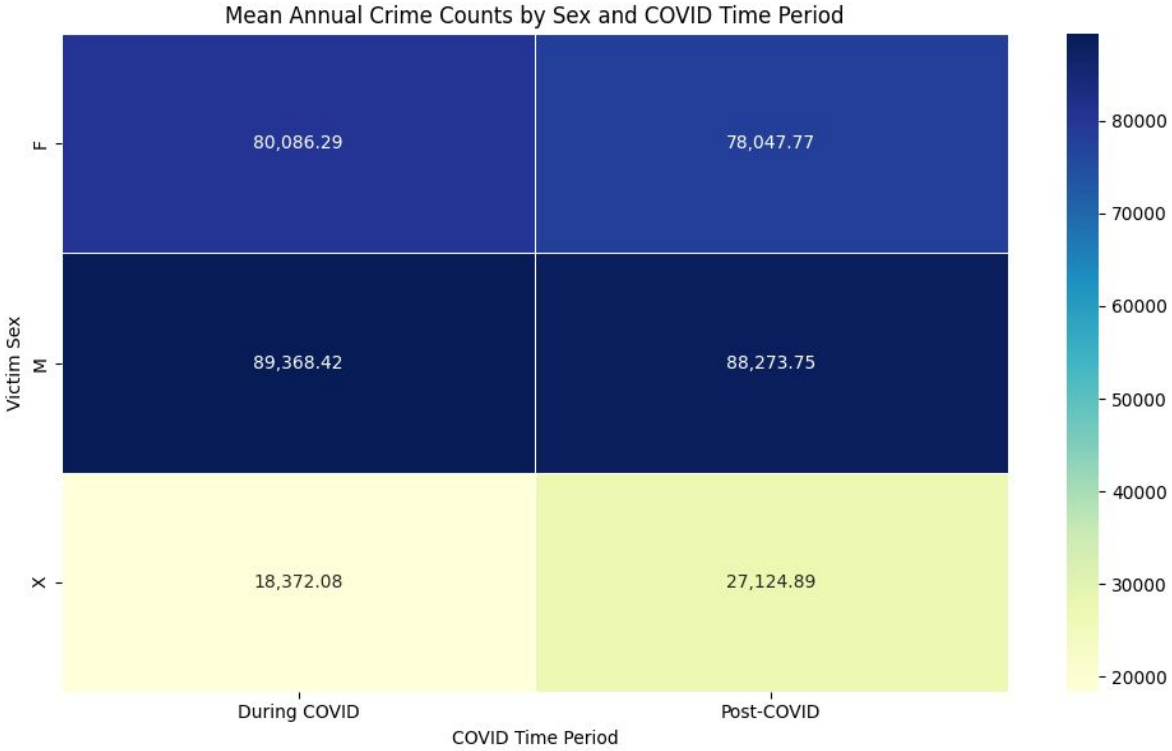
Question 3: How did crime against certain victim demographics change during COVID versus post- COVID?



Increase by 25% of victimless crimes!

Also, increase in crimes against young adults (13-18) and the elderly (66+).

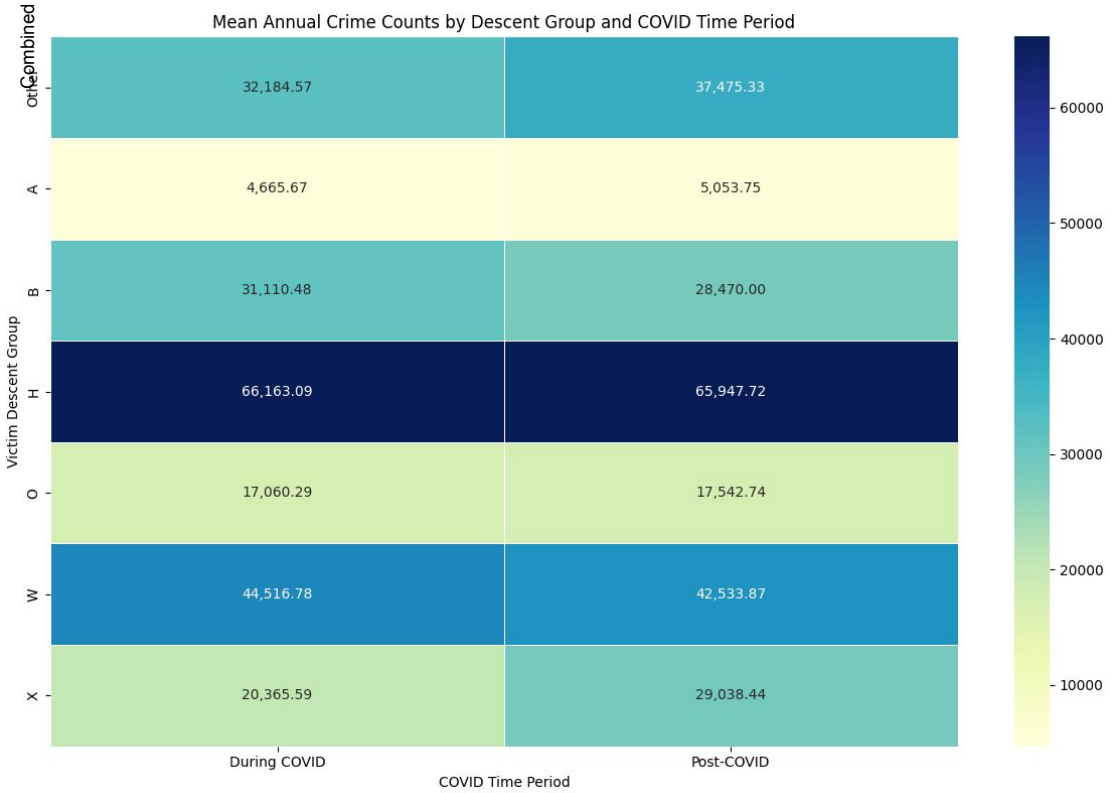
Question 3: How did crime against certain victim demographics change during COVID versus post- COVID?



Increase by 46% of crimes with unknown sex, some of which answered by victimless crimes.

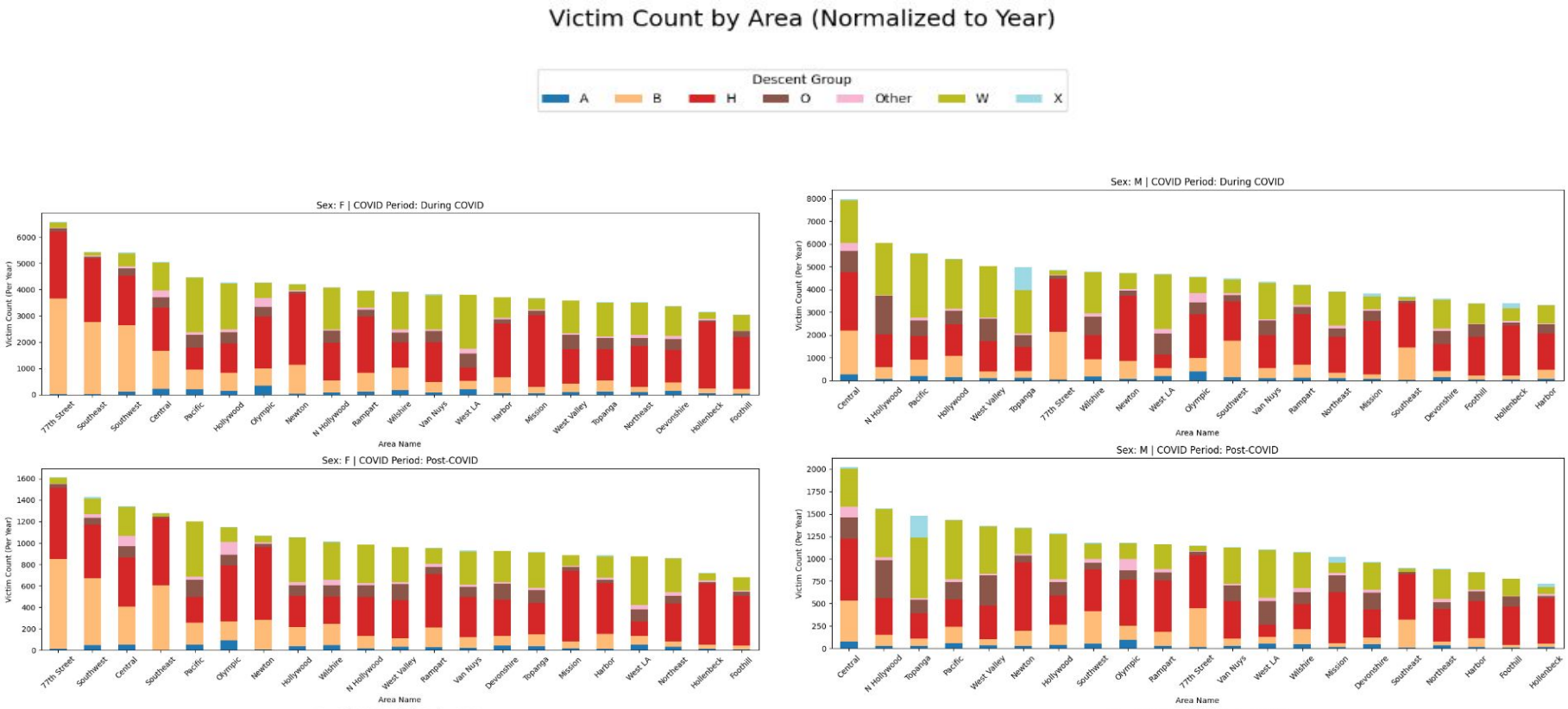
Miniscule decrease to crime against male and female sex.

Question 3: How did crime against certain victim demographics change during COVID versus post- COVID?



Increase of crime by 8.3% against other Asians (Not Chinese, Japanese, Cambodian, Filipino, or Asian Indian).

42.6% increase in crimes where descent unknown, some of which answered by victimless crimes.



Most importantly, for females there seems to be a higher rate of hispanic victims in the top five areas of crime counts, but for men there seems to be a higher rate of white victims in the top five areas of crime counts.

Conclusion

1. **H1: Crime changed from COVID to Post COVID and also crime amount increased by area (Somewhat True)**
   a. Only a half of the areas crime percentage increased in post–COVID.
   b. Not all the areas crime major crime type changed, most of them remain the same except 5 areas

2. **H2: Crime changed in frequency, type, and time between the two periods (mostly false)**
   a. COVID-19 did result in an initial dive in some types of Crime
   b. As the pandemic wore on, most crimes increased in frequency
   c. Some crimes varied in time of execution between the two periods (Homicide)

3. **H3: Crime against certain victim demographics changed during these periods (True)**
   a. Post - COVID, we see a trend of higher victimless crimes (e.g. traffic violations and theft).
   b. Post - COVID, We see an increase of crime against those in the other Asian category.
   c. There was a miniscule decrease in crime against men and women (between 1 to 2.5%).
   d. For females, there is a higher rate of hispanic victims in the top five crime count areas. For males, there is a higher rate of white victims in the top five crime count areas.